metering
TENGYUE

Product
Manual

TS18 Ultrasonic Water Meter

Innovating Measurement Professional Smart Meter Manufacturer

ISO 4064

OIML

WRAS

MID

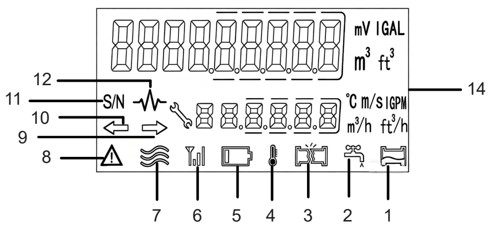
Basic Information

TS8 is an advanced electronic ultrasonic water meter with large caliber and high precision. It is a metering instrument that uses the ultrasonic time-difference method to continuously measure, record, and display the volume of water flowing through the sensor. It can meet the strict requirements of water metering management departments for accurate measurement of water consumption. The TS8 ultrasonic metering water meter is compatible with IoT transmission technology, providing users with different customized solutions. It is widely used in water supply main lines, agricultural irrigation, industrial water use and other fields.

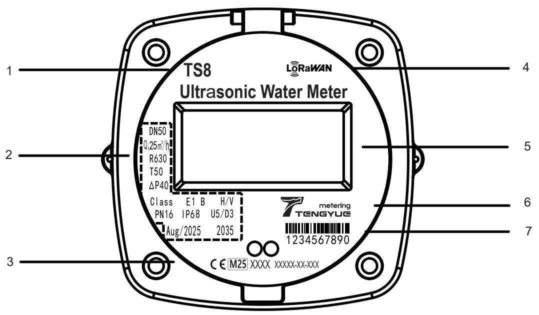
- No moving parts in the measuring mechanism, no wear, and the accuracy will not decrease during the service life of the meter;
- Supports 9-digit LCD display;
- Supports vertical/horizontal or inclined installation angles;
- Supports reverse flow detection for accurate measurement;
- Supports non-full pipe detection;
- Low pressure loss;
- IP68 protection level, adaptable to various harsh installation environments;
- Supports remote reading of meter data: wired reading via RS-485/M-Bus/pulse, wireless reading via LoRa/LoRaWAN/W-MBUS/NB-IoT.

Product Disassembly

1	Non-full Pipe Alarm
2	Drip/Leakage Alarm
3	Burst Pipe Alarm
4	Temperature Indicator
5	Low Pressure Alarm
6	Signal Strength
7	Water Flow Status Indicator
8	Fault/Alarm Indicator
9	Forward Flow Indicator
10	Reverse Flow Indicator
11	Meter Address
12	Signal Amplitude
13	Unit



Nameplate



1 Model
2 Specifications / Meter Parameters
3 Certification Mark
4 Communication Method / Protocol
5 LCD Display Area
6 Manufacturer/Company Logo
7 Meter Address

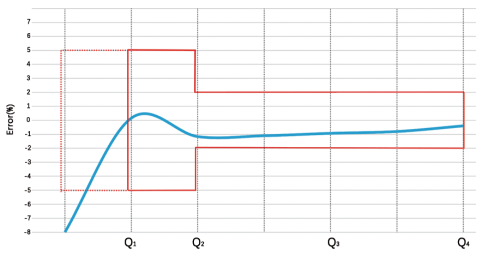
1	DN50	Nominal Diameter
2	Q3 2.5m³/h	Permanent Flow Rate
3	R630	Q3/Q1
4	T50	Temperature Class
5	ΔP40	Pressure Loss Rating
6	Class 2	Accuracy Class
7	E1	Electromagnetic Environment Class
8	B	Environmental Class
9	H/V	Installation Method
10	PN16	Pressure Rating
11	IP68	Protection Rating
12	U5/D3	Flow Sensitivity Class
13	Aug/2025	Manufacturing Date (Month/Year)
14	2035	Service Life

Technical Specifications

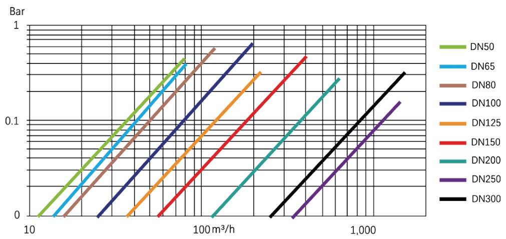
Model	Q4	Q3	Q2	Q1	R(Q3/Q1)	Minimum Flow Rate	Pressure Loss Class
DN	m³/h	m³/h	m³/h	m³/h	/	m³/h	△P(bar)
TS8-50	31.25	25	0.16	0.1	R250	0.0143	0.4
	50	40	0.16	0.1	R400		
TS8-65	50	40	0.256	0.16	R250	0.0229	0.4
	78.75	63	0.256	0.16	R400		
TS8-80	78.75	63	0.4	0.252	R250	0.036	0.25
			0.252	0.16	R400		
TS8W100	125	100	0.64	0.4	R250	0.0571	0.25
			0.4	0.25	R400		
TS8-125	200	160	1.024	0.64	R250	0.0914	0.25
			0.64	0.4	R400		
TS8-150	312.5	250	1.6	1	R250	0.1429	0.25
			1	0.625	R400		
TS8-200	500	400	2.56	1.6	R250	0.2286	0.16
			1.6	1	R400		
TS8-250	787.5	630	4.03	2.52	R250	0.36	0.16
			2.52	1.575	R400		
TS8-300	1250	1000	6.4	4	R250	0.5714	0.16
			4	2.5	R400		
			6.4	4	R400		

Description	Value	Unit
Accuracy Class	2	-
Q3/Q1	R250/R400	-
Maximum Reading	99999999.9	-
Maximum Working Pressure	1.6	MPa
Temperature Class	T30/T50/ (T70 optional)	℃
Installation Condition Sensitivity Class	U10/D5	-
Protection Level	IP68	-
Environmental Class	O	-
Electromagnetic Class	E1	-

Standard Error Curve Chart

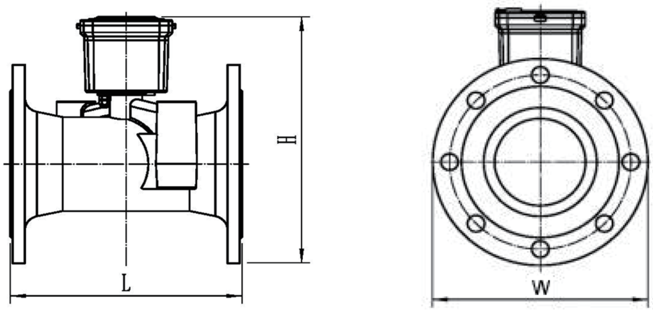


Pressure Loss Curve Chart



Feature	Description		
Mode	**4G (Cat.1)**	**LoRaWAN**	**W-MBUS**
Frequency Band	Full band	EU868/US925/AU923	433MHz/868MHz
Radiated Power	Maximum 23+2 dBm	Maximum 14 dBm	Maximum 14 dBm
Effective Range	5-15 km (open environment)	1-3 km (open environment)	300-500 m (open environment)
Protocol	3GPP	LoRaWAN v.1.0.3	EN13757 OMS
Communication Mode	TCP/UDP	Class A	T1 (default), C1
Power Supply	3.6V DC 8.5Ah	3.6V DC 8.5Ah	3.6V DC 8.5Ah
Upload Interval	24 h	12/24 h	30 min
Corresponding Mode	Timed/Interval	Timed	Timed

Dimensions



Nominal Diameter(mm)		Length L	Width W	Hight H	Flange Connection		
					Flange Diameter	Bolt Circle Diameter	Bolt Size—M
DN50	2 Inch	200	170	215	170	125	4—M16
TS8W-65	2.5 Inch	200	185	220	185	145	4—M16
TS8W-80	3 Inch	225	200	235	200	160	8—M16
TS8W-100	4 Inch	250	220	255	220	180	8—M16
TS8W-125	5 Inch	250	250	285	250	210	8—M16
TS8W-150	6 Inch	300	285	335	285	240	8—M20
TS8W-200	8 Inch	350	340	405	340	295	12—M20
TS8W-250	10 Inch	450	405	490	405	355	12—M24
TS8W-300	12 Inch	500	460	550	460	410	12—M24

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